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Reading the Fed with a local LLM

Tone, the 2-year yield, and look-ahead bias

Hadrien Tramoni

Code, data pipeline and full protocol: github.com/HadrienT/text-signals

Abstract. A local 14 B open-weight model (Phi-4) reads each FOMC statement from 1994 to 2026 into a structured extraction with verbatim evidence. Its aggregate tone score explains the 2-year Treasury yield's move in the 30 minutes around the release, beyond the rate surprise and two standard text baselines. A one-standard-deviation more hawkish statement comes with a +1.03 bp move (HAC $p = 0.004$, $n = 243$), under a specification fixed before any score met market data. The effect survives every pre-specified robustness check and a permutation placebo, and adds 2.1% out-of-sample R^2 over the baseline. It is also narrow: it does not reach the daily close, it is weak on press conference days, and its out-of-sample gain was earned before 2009 and in 2022. Because the model has read the history of almost every statement, look-ahead bias is tested from several angles. The decisive one rewrites each statement's forward guidance: the score follows the text every time, as much for statements the model may remember as for those it cannot. A frontier model gives the same signal with cleaner extraction. A retrieval agent built on the same corpus answers most questions correctly with verbatim citations, but it also shows how such citations can mislead.

1 Question and design

When the Federal Open Market Committee releases a statement, the rate decision is usually priced in. What moves bonds is what the text implies about the future path of policy (Gürkaynak et al., 2005). This paper asks two questions. Can an LLM reading only the text measure that stance well enough to explain the market's reaction, beyond the decision and beyond dictionary and encoder baselines? And how much of the result survives once look-ahead bias is addressed?



Figure 1. The pipeline. Every document is keyed by its release timestamp, and every LLM output is cached with its model, weights, prompt and schema versions.

Event and outcome. There is one observation per statement: the change in the 2-year yield in the 30-minute window around the release, from the San Francisco Fed's event-study database (USMPD; Acosta et al., 2025). The window isolates the statement from the press conference and same-day news; the daily change is kept as a robustness outcome. Release dates were checked against four sources, and two calendar errors and four misdated minutes in the intraday database were corrected.

Model. The model is estimated by OLS,

$$\Delta y_i = \alpha + \beta\,\text{tone}_i + \gamma\,\text{MP1}_i + \delta'\,\text{baselines}_i + \varepsilon_i,$$

where MP1 is the futures-based surprise in the rate decision (Kuttner, 2001), defined in Appendix A. The baselines are the Loughran and McDonald (2011) net negative share and the mean $P(\text{neg}) - P(\text{pos})$ of FinBERT (Araci, 2019). Inference uses Newey–West standard errors (Newey and West, 1994) with 4 lags, chosen a priori. There is one primary test ($H_0 : \beta = 0$), and the 14 secondary tests are corrected together by Holm. Power was computed before scoring: about 0.92 bp per SD is detectable with 80% power. The sample has 250 statements; 243 enter the primary test.

2 Measuring tone

The model fills a typed schema: five stance fields on a common −2 (dovish) to +2 (hawkish) scale (forward guidance, inflation, labour market, growth, balance sheet), the risk balance, the decision with its size, and dissents. Every field carries a span copied verbatim from the statement as evidence. The tone score is the equal-weight mean of the filled stance fields, so no weight is fitted to the outcome.

There is no ground truth for the extraction. Hand labelling was stopped after five practice statements, and a stronger model (Claude) served as a blind *reference reader*. Agreement with it measures consistency, not correctness. Two checks need no reader: the decision and its size against FRED, and whether each evidence span appears verbatim. On a fresh confirmation sample, fixed in advance, Phi-4 gave 0 invalid outputs, a tone correlation of 0.86 with the reference and a forward guidance κ of 0.62. Rejected choices are reported too: a model that inverted the sign of balance-sheet purchases ($\kappa = 0.07$), an evidence-by-sentence schema that collapsed the forward guidance κ (0.84 to 0.30), and a repair loop that changed values. In the end, scores use the model's first answer only.

3 Results

3.1 Primary test

Regressor	β per SD (bp)	HAC p
LLM tone (Phi-4)	**+1.03**	**0.004**
MP1 (rate surprise)	+2.11	0.002
Loughran–McDonald net negative	−0.39	0.28
FinBERT net negative	−0.00	0.99

Table 1. Primary test, 30-minute change in the 2-year yield, $n = 243$. R^2 rises from 0.198 to 0.221 when the LLM score is added; the baselines carry nothing once it is in.

Robustness check (all fixed in advance)	β per SD (bp)	p
HC3 standard errors	1.03	0.014
Moving-block bootstrap, 95% interval	1.03	[0.36, 1.77]
Without the 19 influential statements (Cook's $4/T$)	1.01	0.001
Huber robust regression	0.73	0.022

Table 2. Robustness. Residuals are heteroskedastic and fat-tailed but not autocorrelated, and collinearity is negligible (VIF ≤ 1.63).

3.2 Where the effect stops

Five of the 14 secondary tests survive Holm. The ones that bound the result:

- **No effect on the daily close** (−0.07 bp per SD, $p = 0.91$). The tone moves yields inside the window. Nothing here shows the move lasts: it may be reversed or swamped later in the day, or the daily change may be too noisy.
- **Much weaker on press conference days**: +0.48 ($n = 96$) against +1.70 on other days. When the Chair speaks 30 minutes later, markets seem to wait, and since 2019 that is every meeting.
- The change in tone from the previous statement points the same way (+0.85) but does not survive correction (Holm $p = 0.079$).

3.3 Out of sample and placebos

The model is refitted every January on past statements only and forecasts that year, from 2000 to 2026. Adding the LLM score lowers the squared forecast error by **2.1%** against the baseline; the Clark–West test of equal accuracy rejects with one-sided $p = 0.003$ (Clark and West, 2007). The gain is not steady (Figure 2).

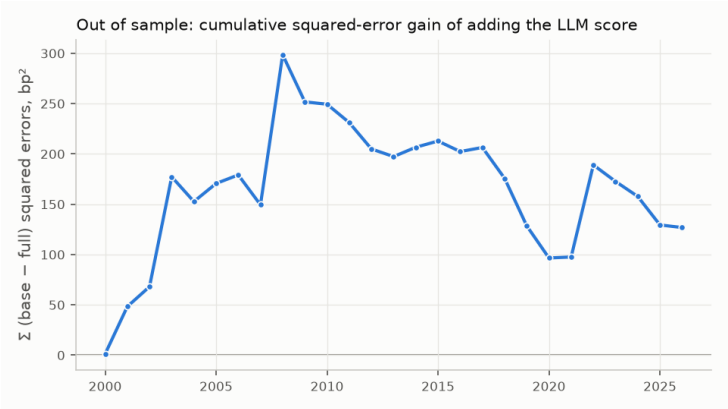


Figure 2. Cumulative out-of-sample squared-error gain of adding the LLM score. It is earned in 2000–2008 and in 2022, and lost in most other years; on recent years alone, the score would not have helped a forecaster.

Shuffling the score across statements (9,999 permutations) leaves t-statistics centred on zero, and the observed t of 2.85 lies beyond 99.5% of them. The previous statement's score explains nothing on its own. One placebo expectation failed. Placed next to the current score, the previous one is significantly negative (-1.04, $p = 0.034$), as if the market priced the new tone net of the tone already known. That reading was formed after the result and is not tested.

4 Look-ahead bias

Phi-4's training data end in June 2024, so it has read the history of 230 of the 250 statements. A model that knows how markets took a statement could let that shade its reading.

Test	Result
Cutoff split	$\beta = -0.44$ (se 1.38) on the 19 statements after the cutoff: not detectable; no power, stated in advance.
Regime-matched split	On press conference days before the cutoff, $\beta = +0.56$ ($n = 77$): inconclusive by the pre-set rule. The regime halves the effect but does not remove it.
Anonymisation	With dates and names removed, scores correlate 0.96 with the originals; β is 0.85 against 0.97.
Dating probe	The model dates anonymised statements: 81% within one year, but it never places a statement after 2023. It knows the period.
Look-ahead-free reader	Models trained only on earlier text (ChronoGPT vintages, He et al., 2025) cannot read the statements ($r = 0.04$ against a 0.30 bar): no evidence either way.
Counterfactual edits	**On 38 statements, a hawkish rewrite of the guidance sentence beats the dovish one 100% of the time, before and after the cutoff alike.** Statements the model may remember move slightly more (+0.33, 95% interval −0.07 to +0.68).

Table 3. Look-ahead tests. The edits were written and frozen before any edited text was read.

Look-ahead needs two steps: the model knows the period, and that knowledge shades the score. The probe shows the first. The counterfactual edits test the second directly and do not find it. Two statements, one on

each side of the cutoff, share the same guidance sentence and got identical rewrites, and the model reads them identically. Statements whose text kept other guidance sentences, which would pull the reading back as memory would, were flagged in advance, and the result holds without them. Look-ahead is not excluded, since memory could still shade borderline readings where text and memory agree, but nothing found indicates that it drives the result.

5 Does a frontier model read better?

As an exploratory second reader, specified before any reading, Claude Opus scored all 250 statements. For economy it read them in batches rather than one per call as the local model did, on anonymised texts shuffled across years to limit recognition.

	Phi-4, original	Phi-4, anonymised	Claude, anonymised
Decision right (FRED)	–	96%	96%
Statements with non-verbatim evidence	–	~33%	**0%**
β per SD (bp), HAC p	0.97, 0.006	0.84, 0.012	0.90, 0.007
Out-of-sample R^2 vs baseline	2.6%	1.3%	0.4%

Table 4. Same 239 statements throughout. The two tone scores correlate at 0.89, and in a joint regression neither adds to the other.

The frontier model extracts more cleanly but finds no stronger signal. The tone of a statement is coarse enough that a 14 B local model captures what moves the 2-year yield. The comparison also gives an indirect argument on look-ahead. Claude's training data cover all but three statements, so if knowing what followed drove the scores, the model that knows most should fit best, and it does not.

6 A retrieval agent, and how far its citations can be trusted

A separate question is reliability: when an LLM answers a research question and cites its sources, can the citations be trusted? A question-answering agent over the corpus (250 statements, 262 sets of minutes, 96 press conference transcripts; 26,354 passages) combines BM25 and a dense encoder by reciprocal rank fusion, with an `as_of` filter for the point-in-time rule. Phi-4 answers from the top eight passages with verbatim quotes, or refuses, and code checks every quote. Fifty questions with reference passages were written before any answer, split into 10 for development and 40 for a single test run; ten were traps the agent should refuse.

Test set, 40 questions	Result
Retrieval recall@8 (fusion; BM25 0.50, encoder 0.53)	0.59
Answerable: right / partial / wrong / refused (of 32)	21 / 6 / 1 / 4
Traps refused	6 of 8
Citations verbatim (none invented)	60 of 63

Table 5. Indexing each passage with a header from its document's metadata, a change made on the development questions, raised recall@8 from 0.25 to 0.59 on the test set.

Three lessons apply to any such tool. **Retrieval is the bottleneck**: when the right passage is retrieved, the answer is almost always right. **Citations are reliable word for word, but not in time**: two answers quote exact text from the wrong meeting, attributing a June 2022 dissent to March 2022, and checking the quote alone would not catch it. **A point-in-time filter is not enough**: asked "as of" 1 December 2015, before the first rate rise, the agent gave the correct range, which appears in none of the retrieved passages.

It answered from memory, against its instructions: look-ahead in miniature.

7 Limitations

- **Small sample, wide intervals.** With 243 events, the estimate is close to the minimum detectable effect.
- **Narrow effect.** It is intraday only and weak when a press conference follows, which limits its relevance to today's FOMC.
- **No ground truth** for the extraction; the hardest field, forward guidance, has a κ of 0.6–0.7 out of sample.
- The intraday data are taken from USMPD as published.
- **Negative and failed results** are reported in the protocol: the daily outcome, the change in tone, a failed placebo expectation, a look-ahead-free reader too weak to use, an inconclusive regime test, rejected extraction variants and an imperfect trap question.
- **Labelled post-hoc choices.** Choices made after seeing a result are labelled as such, and each was made on development data before a single test run.

8 Implications and next step

For investment research, a local open-weight model gives an auditable, verbatim-sourced reading of central bank text with a measurable link to rates. It costs nothing at the margin, and no document leaves the machine; for this task, a frontier model buys auditability, not signal. The link is real but narrow, so it is one input to a rates view, not a strategy. Look-ahead bias needs experiments of its own. Here, cutoff splits had no power and anonymisation removed too little, and editing the text was what separated reading from remembering.

The two halves of the work can be joined. The failed placebo suggests that markets respond to tone *relative to what was already known*. The retrieval agent can measure this directly. With the `as_of` filter set to the day before each release, it can retrieve the previous statement, the latest minutes and the latest press conference, and ask the model what the new statement changes, with evidence on both sides. That score is point-in-time by construction. Because the idea came from a result already seen, it would have to be specified and frozen before computing it, and reported as exploratory.

Reproducibility. Everything a result depends on lives in the repository and is tested. Raw files are stored with URL, fetch time and sha256, and every LLM output with model, weights, prompt and schema versions. `just results` recomputes every number in this paper from the data and caches. The protocol records each decision with its date and reason.

Data and software

- FOMC statements, minutes and press conference transcripts: Board of Governors of the Federal Reserve System, federalreserve.gov/monetarypolicy/fomccalendars.htm.
- Intraday yield changes and the MP1 surprise: U.S. Monetary Policy Event-Study Database, Federal Reserve Bank of San Francisco, sffed.us/usmpd.
- Daily Treasury yields and the federal funds target: FRED, Federal Reserve Bank of St. Louis, fred.stlouisfed.org.
- Loughran–McDonald master dictionary: sraf.nd.edu/loughranmcdonald-master-dictionary.
- Phi-4, Q8_0 GGUF weights: huggingface.co/microsoft/phi-4-gguf.
- FinBERT: huggingface.co/ProsusAI/finbert. Retrieval encoder bge-small-en-v1.5: huggingface.co/BAAI/bge-small-en-v1.5. ChronoGPT vintages: huggingface.co/manelalab.
- Code, protocol and data pipeline: github.com/HadrienT/text-signals.

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A The MP1 surprise

MP1 measures how much the rate decision itself surprised the market, using fed funds futures (Kuttner, 2001). A fed funds futures contract for a given month settles on the *average* effective federal funds rate over that month, and its implied rate is 100 − price. Its change in a narrow window around a release therefore reveals how the market's expectation for the month moved when the decision became known.

Let Δf^1 be the change in the implied rate of the current-month contract between 10 minutes before and 20 minutes after the release, D the number of days in the month and d the day of the meeting. A change in the rate decided on day d only affects the remaining $D - d$ days of the monthly average, so the surprise in the rate itself is

$$\text{MP1} = \frac{D}{D-d}\,\Delta f^1.$$

When the meeting falls in the last seven days of the month, the scaling factor becomes large and noisy, and the change in the next month's contract, Δf^2, is used instead.

Example. For a meeting on the 16th of a 30-day month, a 5 bp rise in the current-month implied rate means that the decided rate was $5 \times 30/14 \approx 10.7$ bp above what the market expected.

Source and checks. The values come from the USMPD (Acosta et al., 2025). We recomputed MP1 from the database's futures changes and matched it on all 280 statements. In the primary sample, its standard deviation is 6.9 bp.

Why MP1 and not a broader surprise. The database also provides a surprise built from futures over about a year (Nakamura and Steinsson, 2018). That measure includes the market's reaction to forward guidance, the very content the text score is meant to capture, so using it as a control would absorb the effect under study. MP1 removes only the surprise in the current decision.